

# Instrument Recognition – Timbre Analysis

AI in Music, Winter Term 2025/2026

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## Exercise 1 [ADSR-Envelope, 2 points]

Download the guitar and piano samples provided on Moodle. Plot the waveforms of `piano_1.wav` and `guitar_1.wav` and roughly visualise their Attack-Decay-Sustain-Release envelopes.

## Exercise 2 [Overtone Analysis, 4 points]

An important timbre characteristic of instruments is the composition of overtones. For the four files you have been given, analyse the energy distribution across the overtone series for the first eight overtones (Note: the pitch is an F4 which lays at 349.23 Hz). What are the differences in overtone composition between the piano and guitar samples?

The difference between `guitar_1.wav` and `guitar_2.wav`, as well as the piano samples, is that different “sound bodies” were used. This could mean a different manufacturer or model, for example. Are there discernable differences caused by the use of a different sound body with regards to the overtone composition?

## Exercise 3 [Noise Analysis, 4 points]

The attack phase of an instrument is a noisy part of the signal that contains important characteristics of the instrument. Isolate only the first 100 ms of each sample. Then, make Q-Q plots (`sklearn.stats.probplot`) of these short waveforms to visually gauge whether the noise is normally distributed. The noise pattern of which instrument is closer to a normal distribution? Hypothesise as to why this may be the case by listening closely to the attack phase of each sample.